

Bridging the Gap Between Plausibility and Admissibility: Constraint-Aware Flow Maps for Dynamic Graph Systems

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Abstract

Generative models can support decision-making under uncertainty by producing ensembles of plausible future system trajectories, but statistical plausibility does not ensure structural feasibility. This study investigates whether post-sampling symbolic constraints can improve the reliability of generative trajectory modeling in dynamic graph-structured systems. A conditional diffusion model generates future graph-state trajectories from partial observations, while an external symbolic layer applies hard filtering, soft weighting, or projection-based repair. The framework is evaluated on two controlled synthetic regimes: a compact graph and a medium-complexity dependency graph, using metrics for structural validity, sample efficiency, diversity, robustness, and calibration. In the compact regime, the model produces an invalid probability mass of 0.002996, indicating an almost entirely admissible trajectory manifold. Under the same architecture and training protocol, invalid mass increases to 0.155929 in the medium-complexity regime. Hard filtering removes all invalid retained trajectories while preserving 84.4% of generated samples, whereas soft weighting preserves effective sample size but yields only limited validity gains. Family-level analysis shows that dependency constraints account for nearly all observed inadmissibility. These results indicate that statistical plausibility and structural admissibility are distinct reliability properties and that symbolic constraint handling becomes more valuable as graph-structural complexity increases.

Keywords: flow map learning; transport-based generative modeling; conditional diffusion; constraint-aware generative modeling; dynamic graphs; generative trajectory modeling; structural admissibility; reliable decision-making; neurosymbolic AI

1 Introduction

Many decision-support problems involve systems whose state evolves through interconnected components. Examples include cyber-physical infrastructures, communication networks, logistics processes, sensing systems, and operational environments that can be naturally represented as dynamic graphs, in which nodes represent system entities and edges encode dependency relations [1, 2]. Such systems may also exhibit multilayer and interdependent structures, in which local state changes propagate through coupled dependencies and produce system-level effects that are not captured by component-wise analysis alone [3, 4].

In these settings, decisions must be made under uncertainty. Rather than predicting a single future state, it is often preferable to reason over an ensemble of possible future trajectories

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generated from partial observations [5, 6]. Recent advances in diffusion models, score-based generative modeling, flow-based methods, and neural dynamical systems provide powerful tools for learning such conditional trajectory distributions [7–13].

However, statistical plausibility alone does not guarantee decision-making reliability. A generated trajectory can appear realistic while violating structural requirements that define system feasibility. Typical examples include inconsistent dependency propagation, invalid recovery sequences, or violations of operational constraints. This motivates a distinction between three complementary reliability dimensions: statistical plausibility, structural admissibility, and decision-relevant probability calibration [5, 14–16].

This paper investigates whether an external symbolic constraint layer can improve the reliability of generative trajectory modeling in dynamic graph-structured systems. We adopt a flow-map perspective in which a neural generator maps latent uncertainty and partial observations into future graph-state trajectories:

$$\tau = T_\theta(z, O_t). \quad (1)$$

Throughout this paper, the term *flow map* is used in the dynamical-systems sense to denote the learned conditional operator that maps partial observations and latent uncertainty into future graph-state trajectories.

In the diffusion-based instantiation used below, z denotes the full source of randomness for sampling, including the initial diffusion noise and, when applicable, any stochasticity introduced during reverse sampling. Thus, T_θ should be read as the induced conditional sampling map rather than as a claim that all reverse-sampling procedures are deterministic.

The generator is instantiated using a conditional diffusion model, while structural admissibility is evaluated only after sampling, using symbolic constraints. This separation enables us to isolate the contribution of post-sampling constraint handling without modifying the underlying generative model. More generally, constraints can be incorporated during training, used to guide the sampling process, or applied after generation through rejection, reweighting, or repair mechanisms [17, 18]. The present study focuses on the latter setting.

Generative-model evaluation, graph prediction, constrained generation, and probabilistic calibration emphasize different reliability criteria [2, 5, 18, 19]; controlled benchmarks that examine these dimensions jointly for conditional graph-trajectory generation remain limited. We therefore introduce a compact and a medium-complexity regime to assess structural admissibility, sample efficiency, diversity, robustness, and decision-relevant calibration, and to determine how these properties change with graph complexity.

The contributions of the paper are:

1. a constraint-aware flow-map formulation for generative decision support in dynamic graph-structured systems;
2. a conditional diffusion framework for future graph-trajectory generation from partial observations;
3. an external symbolic constraint layer supporting hard filtering, soft weighting, and projection-based repair;
4. a controlled two-regime benchmark for studying complexity-dependent admissibility effects;
5. a reliability evaluation framework that jointly assesses validity, sample efficiency, diversity, robustness, and calibration.

The remainder of the paper is organized as follows. Section 2 introduces the proposed framework and benchmark. Section 3 presents the empirical evaluation. Section 4 discusses implications and limitations. Section 5 concludes the paper.

2 Materials and Methods

2.1 Framework Overview

The proposed framework combines three components: a neural trajectory generator, a symbolic admissibility layer, and a decision layer. Given a partial observation history O_t and latent uncertainty z , the generator produces future graph-state trajectories according to the conditional map introduced in Equation (1). The generator is implemented through a conditional diffusion model that approximates

$$P_\theta(\tau \mid O_t). \quad (2)$$

Generated trajectories are subsequently evaluated by an external symbolic layer through a Boolean validity indicator $I_C(\tau)$ and a continuous violation score $\phi_C(\tau)$. Constraint handling can be performed through hard filtering, soft weighting, or projection-based repair. The resulting trajectory ensemble is then used to estimate probabilities of decision-relevant outcomes.

The architecture is summarized in Figure 1. The explicit separation between neural generation and symbolic validation allows statistical plausibility, structural admissibility, and probabilistic calibration to be analyzed independently.

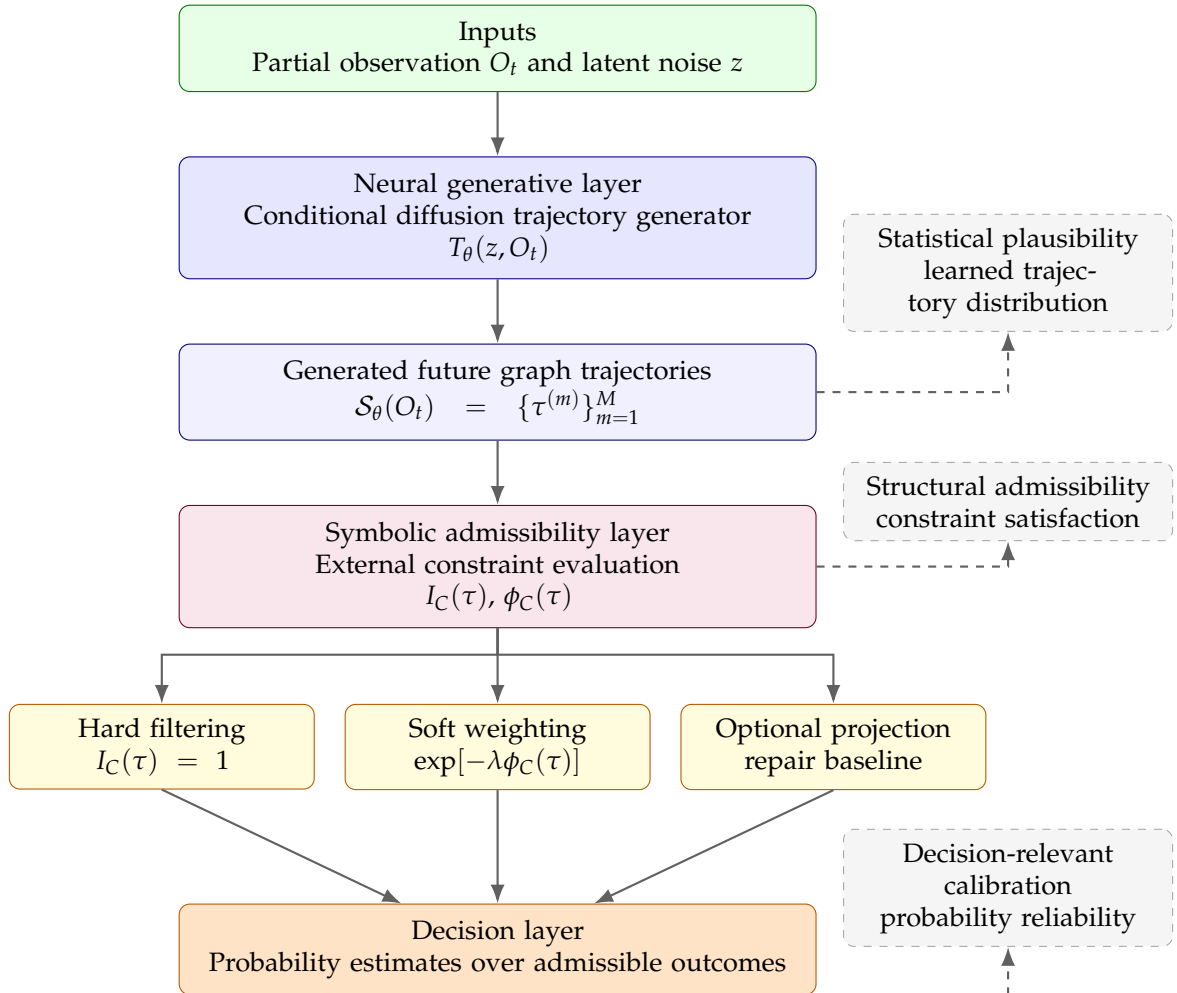


Figure 1: Constraint-aware conditional diffusion framework. Partial observations and latent noise are mapped into future graph-state trajectories, which are evaluated by an external symbolic layer before decision-relevant probability estimation.

2.2 Dynamic Graph Formulation

The system is represented as a dynamic directed graph

$$G_t = (V, E, X_t, A_t), \quad (3)$$

where $V = \{1, \dots, n\}$ is the node set, $E \subseteq V \times V$ is the set of allowed directed dependency edges, $X_t \in [0, 1]^n$ contains node states, and $A_t \in [0, 1]^{|E|}$ contains allowed-edge states. Node values represent operational, functional, or integrity levels, while edge values represent the state of admissible dependencies. Forbidden edges are not generated as independent variables.

The graph-state vector is

$$s_t = [X_t, A_t] \in [0, 1]^d, \quad d = n + |E|, \quad (4)$$

and the available observation history is

$$O_t = \{\tilde{s}_{t-W+1}, \dots, \tilde{s}_t\}, \quad (5)$$

where $\tilde{s}_t$ may be noisy or partially observed. The prediction target is

$$\tau_{t+1:t+H} = \{s_{t+1}, \dots, s_{t+H}\} \in [0, 1]^{H \times d}, \quad (6)$$

which is denoted simply by τ throughout the paper.

In the medium-complexity regime, K_1 represents an intermediate local service outcome, whereas K_2 represents the final global service outcome. The main decision-relevant event is the admissible high-service outcome associated with K_2 , whose probability is estimated from generated trajectory ensembles rather than from single forecasts. Allowed-edge variables are generated jointly with node states and evaluated through boundedness and continuity-related invariants, while dependency constraints are defined at the node level.

2.3 Conditional Diffusion Trajectory Generation

A lightweight conditional diffusion model is used to model $P_\theta(\tau \mid O_t)$. Training follows the standard noise-prediction objective

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{\tau, O_t, \epsilon, k} \left[\|\epsilon - \epsilon_\theta(\tau_k, k, O_t)\|_2^2 \right], \quad (7)$$

where $\epsilon \sim \mathcal{N}(0, I)$ and

$$\tau_k = \sqrt{\bar{\alpha}_k} \tau + \sqrt{1 - \bar{\alpha}_k} \epsilon, \quad (8)$$

with $\bar{\alpha}_k$ denoting the cumulative noise-schedule coefficient.

At inference time, the model generates an ensemble

$$\mathcal{S}_\theta(O_t) = \{\tau^{(1)}, \dots, \tau^{(M)}\}, \quad \tau^{(m)} \sim P_\theta(\tau \mid O_t). \quad (9)$$

The generator receives observations, masks, and diffusion-step embeddings but never receives symbolic constraints, validity labels, or violation scores. Generated continuous outputs are mapped component-wise to $[0, 1]$ before constraint evaluation.

2.4 Symbolic Constraint Layer

Let C denote the set of structural constraints. Each generated trajectory receives a Boolean validity indicator

$$I_C(\tau) = \begin{cases} 1, & \text{if } \tau \text{ satisfies all constraints in } C, \\ 0, & \text{otherwise,} \end{cases} \quad (10)$$

and a continuous violation score

$$\phi_C(\tau) \geq 0, \quad (11)$$

where $\phi_C(\tau) = 0$ for fully admissible trajectories and larger values indicate more severe violations.

Hard filtering retains only admissible trajectories:

$$\mathcal{S}_{\text{hard}}(O_t, C) = \{\tau^{(m)} \in \mathcal{S}_\theta(O_t) : I_C(\tau^{(m)}) = 1\}. \quad (12)$$

When the valid mass is positive, this induces

$$P_{\text{valid}}(\tau \mid O_t, C) = \frac{P_\theta(\tau \mid O_t) I_C(\tau)}{Z_{\text{valid}}(O_t, C)}, \quad Z_{\text{valid}}(O_t, C) = \int P_\theta(\tau \mid O_t) I_C(\tau) d\tau. \quad (13)$$

Soft weighting preserves all samples but downweights trajectories with larger violation scores:

$$w^{(m)} = \frac{\exp[-\lambda \phi_C(\tau^{(m)})]}{\sum_{j=1}^M \exp[-\lambda \phi_C(\tau^{(j)})]}. \quad (14)$$

The corresponding weighted distribution is

$$P_{\text{soft}}(\tau \mid O_t, C) = \frac{P_\theta(\tau \mid O_t) \exp[-\lambda \phi_C(\tau)]}{Z_{\text{soft}}(O_t, C)}, \quad Z_{\text{soft}}(O_t, C) = \int P_\theta(\tau \mid O_t) \exp[-\lambda \phi_C(\tau)] d\tau. \quad (15)$$

This violation-dependent exponential penalty is related to knowledge-based regularization in generative modeling [20], but is applied here only after sampling and therefore does not guarantee structural admissibility.

Projection is included only as a repair-oriented reference baseline. It can be viewed as a rule-based map $\Pi_C : \tau \mapsto \tilde{\tau}$ that clips or adjusts components that violate thresholds, dependencies, or invariant conditions while preserving unchanged components whenever possible.

Five constraint families are evaluated: threshold, dependency, mutual-exclusion, temporal-ordering, and structural-invariant constraints. Families are not mutually exclusive; a trajectory may violate more than one family simultaneously. Full mathematical definitions are reported in Appendix B.

2.5 Benchmark Design

The evaluation uses two synthetic dynamic-graph regimes that share the same generation and evaluation pipeline but differ in graph complexity. Experiment 1 is a compact graph used as a positive-control regime. Experiment 2 is a medium-complexity dependency graph that introduces richer multi-parent dependencies, hierarchical service variables, and temporal constraints.

The benchmark specifications are summarized in Table 1, while the corresponding graph topologies are shown in Figure 2.

Table 1: Summary of the two synthetic dynamic-graph regimes used in the controlled benchmark. The regimes share the same generative and constraint-evaluation protocol, but differ in graph size, state dimensionality, and dependency complexity.

Quantity	Experiment 1 compact	Experiment 2 complex
<i>Graph and target specification</i>		
Nodes	9	15
Allowed directed edges	16	37
Graph-state dimension ($d = n + E $)	25	52
Prediction horizon (H)	16	16
Flattened target dimension ($H \times d$)	400	832
<i>Dataset construction and sampling</i>		
Base trajectories	3000	3000
Trajectory length	80	80
Observation window (W)	8	8
Train/validation/test split	70/15/15	70/15/15
Test contexts per seed	450	450
Samples per context	512	512
<i>Replication</i>		
Random seeds	2026, 2027, 2028	2026, 2027, 2028

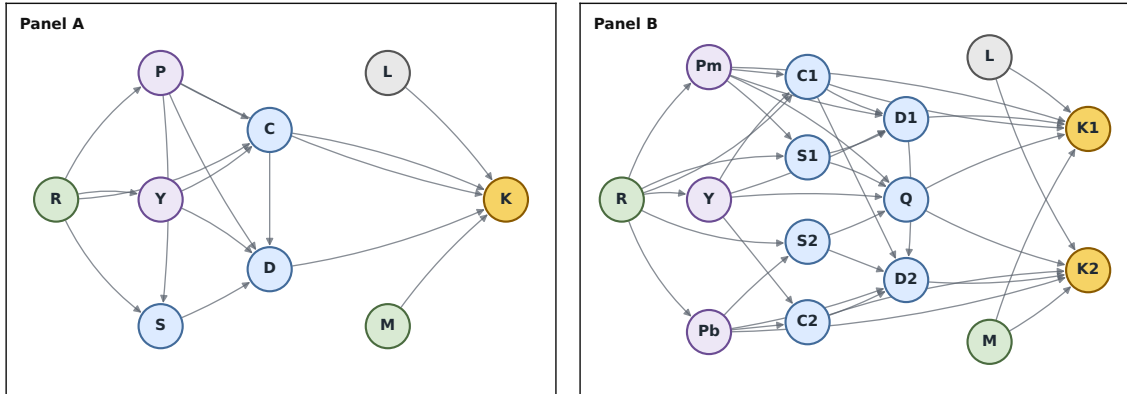


Figure 2: Synthetic dynamic-graph regimes. Panel A shows the compact graph with 9 nodes and 16 allowed directed edges; Panel B shows the medium-complexity dependency graph with 15 nodes and 37 allowed directed edges.

2.5.1 Experiment 1: Compact Graph Regime

Experiment 1 contains nine nodes,

$$V_{\text{compact}} = \{P, Y, C, S, D, L, R, M, K\}, \quad (16)$$

representing power availability, cyber integrity, communication reliability, sensing coverage, data-processing integrity, logistics readiness, repair capacity, maintenance or safe mode, and service feasibility. The graph includes 16 allowed directed dependency edges, yielding $d = 25$ and a flattened target dimension of $H \times d = 400$. This regime tests whether the generator can learn a near-admissible trajectory manifold under limited structural complexity.

2.5.2 Experiment 2: Medium-Complexity Dependency Graph Regime

Experiment 2 contains 15 nodes:

$$V_{\text{complex}} = \{P_m, P_b, Y, C_1, C_2, S_1, S_2, D_1, D_2, Q, L, R, M, K_1, K_2\}. \quad (17)$$

The graph contains 37 allowed directed edges, yielding $d = 52$ and a flattened target dimension of $H \times d = 832$. Compared with Experiment 1, this regime introduces multiple service outputs, richer dependency caps, cross-layer coupling, and more active temporal and cascading constraints. Explicit edge sets and generator settings are reported in Appendix A.

2.6 Dataset, Strategies, and Evaluation

For each regime, 3000 base trajectories of length 80 are generated by a controlled stochastic simulator that evolves node and allowed-edge states under exogenous degradations, recovery dynamics, and cascading dependencies. The simulator is intentionally mostly, but not perfectly, constraint-consistent, avoiding both a trivially valid benchmark and one dominated by structural inconsistency. Observation windows of length $W = 8$ may include noisy node states, noisy edge states, and observation masks, whereas constraint rules, validity indicators, and violation scores are reserved for post-sampling evaluation. Prediction targets have horizon $H = 16$; the data are split into training, validation, and test subsets using fractions of 0.70, 0.15, and 0.15, and results are aggregated across seeds 2026, 2027, and 2028.

Four trajectory-handling strategies are evaluated: unconstrained diffusion, hard filtering, soft weighting, and projection-based repair. They are summarized in Table 2. For soft weighting, the penalty parameter is selected from

$$\lambda \in \{1, 2, 5, 10, 20, 50, 100\}. \quad (18)$$

The selected configuration must preserve a minimum effective sample size ratio to prevent soft weighting from being interpreted as implicit hard rejection.

Table 2: Compared trajectory-handling strategies. Projection is included as an optional deterministic repair baseline.

Strategy	Modifies samples	Modifies weights	Rejects samples	Role
Unconstrained diffusion	No	No	No	Neural generative baseline
Hard filtering	No	No	Yes	Support restriction to admissible trajectories
Soft weighting	No	Yes	No	Conservative probability reweighting
Optional projection	Yes	No	No	Deterministic repair baseline

Structural validity is measured through trajectory-level and step-level constraint-violation rates, invalid probability mass, and violation severity. The trajectory-level violation rate is

$$\text{CVR}_{\text{traj}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{I_C(\tau_i) = 0\}, \quad (19)$$

and the step-level violation rate is

$$\text{CVR}_{\text{step}} = \frac{1}{NH} \sum_{i=1}^N \sum_{h=1}^H \mathbf{1}\{I_C(\tau_{i,h}) = 0\}. \quad (20)$$

For unweighted ensembles, invalid probability mass is

$$\hat{p}_{\text{invalid}} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{I_C(\tau^{(m)}) = 0\}. \quad (21)$$

When computed over the same pooled generated trajectories, [Equations \(19\)](#) and [\(21\)](#) coincide. For soft-weighted ensembles,

$$\hat{p}_{\text{invalid}}^{\text{soft}} = \sum_{m=1}^M w^{(m)} \mathbf{1}\{I_C(\tau^{(m)}) = 0\}. \quad (22)$$

Violation severity is summarized through the continuous score $\phi_C(\tau)$, both globally and by constraint family.

For hard filtering, sample efficiency is measured by the retained ratio

$$r_{\text{hard}} = \frac{|\mathcal{S}_{\text{hard}}(O_t, C)|}{|\mathcal{S}_{\theta}(O_t)|}. \quad (23)$$

The zero-retained-context rate is monitored and reported when nonzero or diagnostically relevant. For soft weighting, sample efficiency is measured through

$$\text{ESS} = \frac{1}{\sum_{m=1}^M (w^{(m)})^2}, \quad (24)$$

and

$$\text{ESS}_{\text{ratio}} = \frac{\text{ESS}}{M}. \quad (25)$$

When reported for hard filtering, $\text{ESS}_{\text{hard}}/M = |\mathcal{S}_{\text{hard}}|/M$ is under uniform weights over retained samples.

Trajectory diversity is computed over the generated ensemble to assess whether constraint handling reduces the variability of possible futures; unweighted or normalized-weighted estimates are used according to the trajectory-handling strategy.

Decision-relevant calibration is evaluated for regime-specific admissible high-service events using the Brier score and Expected Calibration Error (ECE) [\[14, 15\]](#). Let $Y_i \in \{0, 1\}$ denote the realized event for test context i , and let $\hat{p}_i$ denote its probability estimated from the generated trajectory ensemble. The Brier score is

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - Y_i)^2. \quad (26)$$

Calibration is further evaluated as

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|, \quad (27)$$

using equally spaced bins over $[0, 1]$; empty bins do not contribute.

Robustness is evaluated diagnostically, without retraining, under increasing observation noise and missingness, and under topology perturbations that progressively reduce the availability of allowed edges. Structural validity, sample efficiency, diversity, and calibration are recomputed for each severity level. These controlled tests assess degradation and its structural sources, but do not imply that the constraint layer resolves distribution shift, which remains a distinct uncertainty-reliability problem [\[6, 21\]](#).

2.7 Reproducibility

All results are aggregated over three random seeds: 2026, 2027, and 2028. The implementation preserves generated samples, metrics, figures, configuration files, and execution metadata to support reproducibility. Source code and processed artifacts will be released together with the experimental materials described in Appendix D.

3 Results

3.1 Cross-Regime Reliability Comparison

All results are aggregated over three random seeds (2026–2028), using 450 test contexts and 512 generated trajectories per context. The simulator trajectory-level constraint-violation rate remains within the intended sanity range in both regimes, confirming that neither benchmark is trivially valid nor dominated by simulator-level inconsistency. In the compact regime, the generated invalid mass is lower than the simulator-level violation, suggesting that rare simulator violations are smoothed by the learned conditional trajectory distribution.

Table 3 summarizes the main comparison between the two graph regimes. Moving from the compact to the medium-complexity graph substantially increases unconstrained invalid probability mass, from 0.002996 to 0.155929, while reducing the hard-filter retained ratio from 0.997004 to 0.844071.

The contrast is visualized in Figure 3. The compact graph serves as a positive control setting in which the diffusion model learns an almost entirely admissible trajectory manifold. In contrast, the richer dependency structure of the complex graph exposes residual structural inadmissibility, making external constraints substantially more relevant.

Soft weighting preserves the effective sample size in both regimes but only slightly reduces the probability mass that is invalid. Calibration metrics are reported separately because structural admissibility and probability calibration represent distinct reliability dimensions.

Table 3: Main cross-experiment comparison. Increasing graph complexity increases the amount of invalid generated probability mass while the simulator remains within the intended sanity range.

Metric	Experiment 1	Experiment 2	Interpretation
	compact	complex	
Nodes / edges	9 / 16	15 / 37	Structural complexity increases.
State dimension / target dimension	25 / 400	52 / 832	Trajectory dimensionality increases.
Simulator trajectory CVR	0.030222	0.050667	Both regimes remain within the sanity range.
Unconstrained invalid mass	0.002996	0.155929	Generated invalidity increases under complexity.
Hard retained ratio	0.997004	0.844071	Hard filtering becomes nontrivial in the complex regime.
Soft invalid mass	0.002715	0.148807	Soft weighting has a modest effect on invalid mass.
Soft ESS ratio	0.999930	0.998764	Soft weighting preserves effective sample size.
Unconstrained diversity	1.030889	1.561000	The complex regime induces broader variation in trajectories.

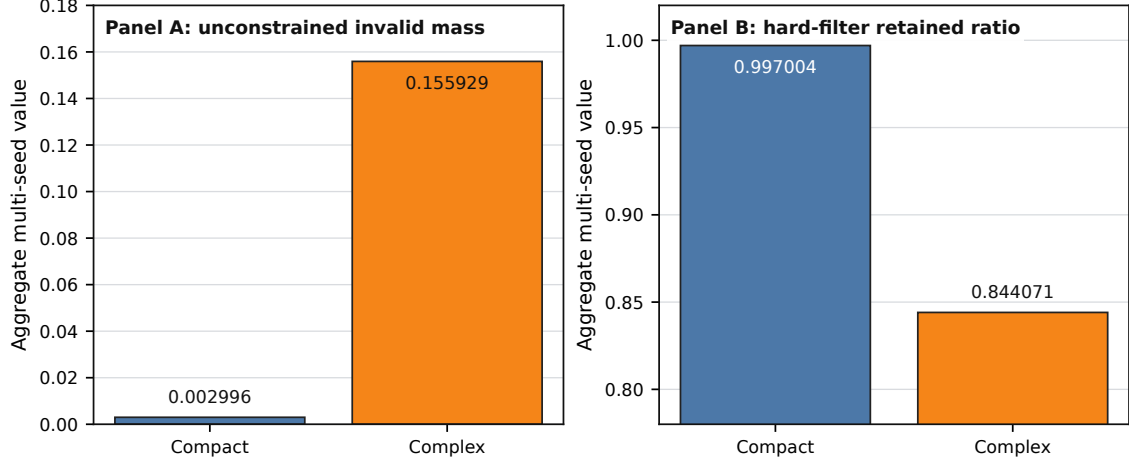


Figure 3: Cross-regime structural effect. Panel A reports the unconstrained invalid-generated mass, while Panel B reports the hard-filter-retained ratio for the compact and medium-complexity regimes.

3.2 Constraint Handling in the Complex Regime

The medium-complexity graph is the primary regime for evaluating post-sampling admissibility mechanisms. Results are reported in Table 4. Hard filtering completely removes invalid retained trajectories, reducing the invalid mass from 0.155929 to 0 while preserving 84.4% of the generated samples. This demonstrates that most generated trajectories remain useful after enforcing all constraints.

Soft weighting produces a different behavior. By construction, it preserves all generated samples and maintains an ESS ratio of 0.998764. However, the invalid mass decreases only from 0.155929 to 0.148807, corresponding to an absolute reduction of 0.007122. These results indicate that soft weighting primarily redistributes probability mass rather than enforcing admissibility.

The projection baseline achieves near-complete validity (0.000023 invalid mass) by modifying generated trajectories and is therefore interpreted as a repair-oriented reference rather than a probabilistic generation strategy. Overall, the three approaches reveal a clear trade-off among validity guarantees, sample preservation, and intervention in generated samples.

Table 4: Strategy-level results in the medium-complexity graph regime. Hard filtering removes invalid retained samples, while soft weighting preserves effective sample size with limited reduction of invalid mass.

Strategy	Invalid mass	Trajectory CVR	Retained ratio	ESS ratio	Diversity	ECE
Unconstrained diffusion	0.155929	0.155929	1.000000	1.000000	1.561000	0.070736
Hard filtering	0.000000	0.000000	0.844071	0.844071	1.539848	0.070643
Soft weighting	0.148807	0.155929	1.000000	0.998764	1.561000	0.070731
Optional projection	0.000023	0.000023	1.000000	1.000000	1.560327	0.070284

3.3 Constraint-Family Diagnostics

Table 5 reports family-level violation diagnostics. In both regimes, dependency constraints are the dominant source of structural inadmissibility. In the medium-complexity graph, dependency invalid mass reaches 0.155929, whereas threshold violations (0.000370) and temporal-ordering violations (0.000383) remain negligible. Mutual-exclusion and structural-invariant violations are absent at the aggregate level.

These results indicate that generated invalidity is not primarily caused by trivial range violations, forbidden-edge artifacts, or maintenance inconsistencies. Instead, it originates from the multi-parent dependency structure intentionally introduced in Experiment 2. Family-level invalid masses are not additive because a trajectory may violate multiple constraint families simultaneously.

Table 5: Per-family constraint diagnostics in the medium-complexity graph regime. Residual invalidity is dominated by dependency-consistency violations, whereas threshold, temporal-ordering, mutual-exclusion, and structural-invariant violations remain negligible.

Constraint family	Invalid mass	Interpretation
Dependency	0.155929	Dominant structural failure mode.
Threshold	0.000370	Negligible relative to dependency violations.
Temporal ordering	0.000383	Negligible relative to dependency violations.
Mutual exclusion	0.000000	No aggregate contribution.
Structural invariant	0.000000	No aggregate contribution.

3.4 Calibration Results

Calibration is evaluated independently from structural validity using the Brier score and Expected Calibration Error (ECE). Across both regimes, constraint handling has only a marginal effect on calibration metrics. In the compact graph, hard filtering slightly changes the Brier score from 0.578380 to 0.577716 and the ECE from 0.741424 to 0.740841. Similar behavior is observed in the medium-complexity graph, where ECE changes from 0.070736 to 0.070643 after filtering.

These findings confirm that improvements in admissibility do not automatically translate into better probability calibration. Structural validity and calibration should therefore be considered complementary but distinct reliability properties. The large numerical difference between calibration scores in the two benchmark regimes should not be interpreted as a direct measure of problem difficulty, since calibration is evaluated with respect to regime-specific admissible-service events and operating ranges.

3.5 Robustness Analysis

Robustness is evaluated under observation noise and topology perturbations without retraining the generator. In the medium-complexity regime, severe observation noise increases invalid mass from 0.154819 to 0.158594 and reduces the hard-filter retained ratio from 0.845181 to 0.841406. Topology perturbations produce a stronger effect, increasing invalid mass to 0.169536 and reducing the retained ratio to 0.830464.

Although the observed degradation remains moderate, the results indicate that topology shifts have a greater impact on structural admissibility than observation-noise perturbations do. Detailed robustness results are reported in Table 12.

3.6 Key Findings

Four findings emerge consistently across the benchmark. First, the diffusion generator learns an almost entirely admissible trajectory manifold on the compact graph, with unconstrained invalid mass of only 0.002996. Second, structural complexity substantially increases generated invalidity: in the medium-complexity regime, invalid mass rises to 0.155929 despite using the same generator architecture and training protocol. Third, hard filtering provides the strongest admissibility guarantee, eliminating invalid retained trajectories while preserving 84.4% of

generated samples. Fourth, dependency constraints account for nearly all observed inadmissibility, while threshold, temporal-ordering, mutual-exclusion, and structural-invariant violations remain negligible.

Collectively, these results show that the benefit of external symbolic constraints grows with graph-structural complexity and that structural validity and probabilistic calibration represent distinct dimensions of reliability.

4 Discussion

4.1 Main Findings

The benchmark highlights a clear complexity-dependent effect in constraint-aware generative trajectory modeling. When the dependency structure is relatively simple, the diffusion generator learns a trajectory manifold that is already almost entirely admissible. In this setting, external constraints primarily act as a certification mechanism.

The situation changes substantially in the medium-complexity graph. Under the same architecture and training protocol, generated trajectories exhibit non-negligible structural inadmissibility. This indicates that increasing graph dimensionality alone is not the main issue; rather, the difficulty lies in capturing higher-order dependency relations, hierarchical service variables, and temporal consistency requirements simultaneously.

The central outcome of the study is, therefore, that the usefulness of symbolic constraint handling increases with structural complexity. As the admissible trajectory manifold becomes more difficult to learn directly from data, post-sampling admissibility mechanisms become increasingly important for reliability.

4.2 Reliability Beyond Distribution Learning

A key observation emerging from the benchmark is that statistical plausibility and structural admissibility are not equivalent. The diffusion model learns realistic trajectory distributions in both benchmark regimes. Nevertheless, in the complex graph, a significant fraction of generated trajectories violates dependency constraints, despite remaining statistically plausible under the learned model. This demonstrates that distribution learning alone does not guarantee structural feasibility.

The results, therefore, support a layered view of reliability. Generated trajectories should be evaluated along at least three complementary dimensions: statistical plausibility, structural admissibility, and decision-relevant calibration. These dimensions are related but not interchangeable. A model may be well calibrated while still generating structurally infeasible trajectories, and conversely, a fully admissible trajectory ensemble does not automatically guarantee calibrated decision probabilities.

4.3 Interpreting Constraint-Handling Strategies

The benchmark clarifies the practical role of the post-sampling strategies investigated. Hard filtering is the only mechanism that provides explicit admissibility guarantees. In the medium-complexity regime, it removes all invalid retained trajectories while preserving a large fraction of the generated ensemble. Its effectiveness depends directly on the amount of invalid probability mass produced by the generator.

Soft weighting addresses a different problem. Rather than enforcing admissibility, it redistributes probability mass away from highly violating trajectories while preserving ensemble continuity and effective sample size. The experiments show that this strategy has only a limited effect on Boolean validity, indicating that probability reweighting and validity enforcement should be regarded as distinct objectives.

Projection occupies a separate methodological position. Because it modifies generated trajectories, it can substantially improve validity, but only by altering the learned generative distribution. It is therefore best viewed as a repair-oriented baseline rather than as a probabilistic inference mechanism. Taken together, the three strategies expose a trade-off among validity guarantees, sample preservation, and fidelity to the original generated distribution.

4.4 Dependency Constraints as the Primary Failure Mode

The family-level diagnostics provide an important insight into the source of the generated inadmissibility. Almost all violations observed in the complex regime originate from dependency constraints. Threshold violations, temporal-ordering violations, mutual-exclusion violations, and structural-invariant violations remain negligible throughout the benchmark.

This result suggests that the primary challenge is not generating realistic local node values but preserving consistency across interconnected system components. In other words, the dominant errors arise from modeling relational dependencies rather than from state-range violations or simple rule violations.

From a decision-support perspective, this distinction is significant. A trajectory may appear plausible when individual state variables are inspected in isolation, yet remain infeasible once dependencies among components are considered. The constraint layer, therefore, provides valuable diagnostic information that cannot be obtained from marginal trajectory statistics alone.

4.5 Implications for Generative Decision Support

The proposed framework can be interpreted as a conditional flow map transporting latent uncertainty into future graph-state trajectories. The symbolic layer subsequently evaluates whether those futures remain structurally feasible.

This separation is particularly relevant in decision-support scenarios. Decision makers rarely require only realistic predictions; they require realistic futures that also satisfy operational, physical, or logical feasibility constraints.

The benchmark demonstrates that the gap between plausibility and feasibility widens as the dependency structure becomes more complex. Consequently, reliability assessment in generative decision support should not rely exclusively on predictive fidelity or likelihood-based criteria. Structural admissibility diagnostics provide complementary information regarding whether sampled futures remain actionable.

4.6 Limitations and Future Work

Several limitations should be acknowledged. First, the benchmark is synthetic and serves as a controlled methodological testbed rather than as an operational validation. Second, only two graph-complexity regimes are considered. Third, constraints are manually specified and assumed to be correct and complete.

In addition, the generator is intentionally prevented from accessing symbolic constraints during training. This design isolates the effect of post-sampling regulation but does not compare against constraint-integrated generative approaches. The observed soft-weighting trade-off is also specific to the adopted violation score and penalty schedule: alternative score designs, adaptive penalties, or guided sampling mechanisms may yield different balances between validity and effective sample size.

Future work should extend the analysis to larger and more heterogeneous graph families, compare post-sampling regulation against constraint-guided generation methods, and investigate settings involving uncertain, incomplete, or dynamically evolving constraints. A particularly promising direction is the integration of admissibility-aware generation with calibration-

aware probabilistic estimation, so that validity and probability reliability can be addressed jointly.

5 Conclusions

This paper investigated constraint-aware generative trajectory modeling in dynamic graph-structured systems from a flow-map perspective. A conditional diffusion model was used to generate future graph-state trajectories from partial observations, while structural admissibility was evaluated via an external symbolic layer that supports hard filtering, soft weighting, and projection-based repair.

The experimental results reveal a clear complexity-dependent effect. In the compact graph, the diffusion model learns an almost entirely admissible trajectory manifold, making external constraints largely diagnostic. In the medium-complexity graph, however, structural inadmissibility becomes substantial despite using the same generation protocol, demonstrating that statistical plausibility does not necessarily imply structural feasibility.

Among the evaluated strategies, hard filtering provides the strongest admissibility guarantees, eliminating invalid trajectories from the retained ensemble while preserving most of the generated samples. Soft weighting preserves sample efficiency but only marginally reduces the probability mass assigned to invalid samples, indicating that probability redistribution and validity enforcement address different objectives. Family-level diagnostics further show that dependency constraints constitute the dominant source of inadmissibility in the generated output.

More broadly, the study supports a layered view of reliability in generative decision support. Statistical plausibility, structural admissibility, and decision-relevant calibration capture distinct properties of generated trajectory ensembles and should be evaluated separately.

Although the benchmark is intentionally synthetic, it provides controlled evidence that the importance of symbolic constraint handling grows with graph-structural complexity. Future work should extend the analysis to larger, more realistic graph environments, compare post-sampling regulation with constraint-guided generation, and investigate methods that jointly address admissibility and probabilistic calibration.

More generally, the results suggest that evaluating generative models solely through predictive plausibility can overlook important forms of structural unreliability. Constraint-aware assessment, therefore, represents a necessary component of trustworthy generative decision support in graph-structured environments.

Author Contributions

Conceptualization: M.R.d.S. and G.L.P.; methodology: M.R.d.S. and G.L.P.; software: M.R.d.S.; validation: M.R.d.S. and A.M.; formal analysis: M.R.d.S. and G.L.P.; investigation: M.R.d.S.; writing—original draft preparation: M.R.d.S.; writing—review and editing: M.R.d.S. and A.M.; supervision: A.M.

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Data Availability

The raw data supporting the conclusions of this article will be made available by the authors on request.

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Conflicts of Interest

The authors declare no conflicts of interest.

A Benchmark Specification

This appendix documents the benchmark configuration used to support the empirical analysis presented in Section 3.

The main paper focuses on how graph-structural complexity affects the relationship between statistical plausibility and structural admissibility. Consequently, only implementation-specific details that are not essential to the main narrative are reported here.

The benchmark consists of two controlled dynamic-graph regimes that share the same generation, sampling, and evaluation pipeline but differ in dependency complexity. Together, they provide the experimental environment used throughout the paper to study reliability in constraint-aware generative trajectory modeling.

A.1 Experiment 1: Compact Graph Regime

The compact-regime node set is introduced in Section 2.5.1. The allowed directed edge set used to instantiate that topology is

$$E_{\text{compact}} = \{P \rightarrow C, P \rightarrow D, P \rightarrow S, Y \rightarrow C, Y \rightarrow D, C \rightarrow D, S \rightarrow D, \\ C \rightarrow K, D \rightarrow K, L \rightarrow K, P \rightarrow K, R \rightarrow P, R \rightarrow C, \\ R \rightarrow Y, R \rightarrow S, M \rightarrow K\}.$$

This edge set provides the compact topology summarized in Table 1 and Figure 2.

A.2 Experiment 2: Medium-Complexity Dependency Graph Regime

The medium-complexity node set is introduced in Section 2.5.2. Table 6 reports the symbol-level interpretation used in the implementation, while the allowed directed edge set is listed below.

Table 6: Node set for the medium-complexity dependency graph.

Symbol	Name	Interpretation
P_m	Main power availability	Primary energy source.
P_b	Backup power availability	Redundant or backup energy source.
Y	Cyber integrity	Integrity of digital control and data channels.
C_1	Internal communication reliability	Local coordination reliability.
C_2	External communication reliability	External coordination or network reliability.
S_1	Primary sensing coverage	Primary monitoring or sensing capability.
S_2	Secondary sensing coverage	Redundant sensing capability.
D_1	Data-processing integrity	First-stage data-processing reliability.
D_2	Data-fusion integrity	Cross-source data-fusion reliability.
Q	Data quality/control integrity	Quality-control and consistency layer.
L	Logistics/resource readiness	Sustained resource availability.
R	Repair/recovery capacity	Capacity to restore degraded components.
M	Maintenance/safe mode	Emergency or maintenance mode restricting service.
K_1	Local service feasibility	Intermediate local decision-relevant service.
K_2	Global service feasibility	Final global decision-relevant service.

The allowed directed edge set used to instantiate the medium-complexity topology is

$$\begin{aligned}
E_{\text{complex}} = \{ & P_m \rightarrow C_1, P_m \rightarrow D_1, P_m \rightarrow S_1, P_m \rightarrow K_1, P_m \rightarrow Q, \\
& P_b \rightarrow C_2, P_b \rightarrow D_2, P_b \rightarrow S_2, P_b \rightarrow K_2, \\
& Y \rightarrow C_1, Y \rightarrow C_2, Y \rightarrow D_1, Y \rightarrow Q, \\
& C_1 \rightarrow D_1, C_1 \rightarrow K_1, C_1 \rightarrow D_2, \\
& C_2 \rightarrow D_2, C_2 \rightarrow K_2, \\
& S_1 \rightarrow D_1, S_1 \rightarrow Q, S_2 \rightarrow D_2, S_2 \rightarrow Q, \\
& D_1 \rightarrow D_2, D_1 \rightarrow K_1, D_2 \rightarrow K_2, \\
& Q \rightarrow K_1, Q \rightarrow K_2, L \rightarrow K_1, L \rightarrow K_2, \\
& R \rightarrow P_m, R \rightarrow P_b, R \rightarrow Y, R \rightarrow C_1, R \rightarrow S_1, R \rightarrow S_2, \\
& M \rightarrow K_1, M \rightarrow K_2 \}.
\end{aligned}$$

This edge set provides the medium-complexity topology summarized in [Table 1](#) and [Figure 2](#).

A.3 Shared Diffusion Configuration

Both benchmark regimes use the same conditional diffusion architecture and training protocol. Keeping the generator fixed across experiments ensures that the observed differences in admissibility can be attributed primarily to graph complexity rather than to differences in model capacity or optimization.

Dataset-level parameters are summarized in [Table 1](#). [Table 7](#) reports the implementation-specific generator configuration used throughout the benchmark.

Table 7: Shared conditional diffusion trajectory generator configuration used in both benchmark regimes.

Component	Configuration
Diffusion steps during training	100
Sampling steps	25
Noise schedule	Cosine
Sampler	DDIM
Prediction type	Noise prediction
Denoiser backbone	Temporal TCN
Hidden dimension	256
Number of layers	4
Dropout	0.10
Time-embedding dimension	64
Conditioning dimension	256
Training objective	Noise-prediction MSE; see Equation (7)
Symbolic constraints provided to generator	No

B Constraint Layer Specification

This appendix formalizes the symbolic admissibility layer introduced in Section 2.4.

Throughout the paper, admissibility is treated as a reliability dimension distinct from both statistical plausibility and probability calibration. The objective of this appendix is therefore to specify how admissibility is operationalized within the benchmark and how the diagnostic quantities used in the empirical evaluation are computed.

Each constraint family contributes to both the trajectory-level validity indicator defined in Equation 10 and the aggregate violation score defined in Equation 11. These quantities support the hard filtering, soft weighting, and diagnostic analyses reported throughout Section 3.

Table 8 summarizes the five constraint families and their associated violation semantics.

Table 8: Constraint families used to evaluate structural admissibility of generated graph trajectories.

Constraint family	Role	Violation score
Threshold constraints	Encode prerequisite conditions for high service or component activation.	Threshold-margin severity.
Dependency constraints	Enforce consistency between downstream variables and upstream parents.	Excess-over-cap severity.
Mutual exclusion	Prevent incompatible states from co-occurring.	Incompatibility severity.
Temporal ordering	Require selected states to recover or activate only after prerequisite stability.	Temporal inconsistency severity.
Structural invariants	Enforce bounded values, allowed-edge consistency, and basic graph-state admissibility.	Invariant-violation severity.

For a generated trajectory τ , each constraint family $f \in \mathcal{F}$ produces a Boolean family indicator $I_f(\tau) \in \{0, 1\}$ and a non-negative family-level violation score $\phi_f(\tau) \geq 0$. The trajectory-level validity indicator introduced in Equation (10) is implemented as the conjunction of all family-level indicators:

$$I_C(\tau) = \prod_{f \in \mathcal{F}} I_f(\tau).$$

The continuous violation score introduced in Equation (11) is implemented as a weighted

aggregation of family-level margins:

$$\phi_C(\tau) = \sum_{f \in \mathcal{F}} \alpha_f \phi_f(\tau), \quad \alpha_f \geq 0.$$

Boolean validity is used for hard filtering and family-level violation rates, whereas the continuous score is used for severity diagnostics and soft weighting. For upper-bound dependency constraints, violation margins are computed as the positive excess above the admissible cap and then aggregated over constraints and time steps.

B.1 Threshold Constraints

Threshold constraints specify prerequisite conditions for high-service outcomes in the medium-complexity regime. For the local service variable K_1 , high activation requires the relevant upstream variables to satisfy their respective threshold conditions:

$$K_{1,t} \geq \theta_{K_1} \Rightarrow P_{m,t} \geq \theta_{P_m} \wedge C_{1,t} \geq \theta_{C_1} \wedge D_{1,t} \geq \theta_{D_1} \\ \wedge Q_t \geq \theta_Q \wedge L_t \geq \theta_L \wedge M_t \leq \theta_M.$$

For the global service variable K_2 , high activation additionally requires support from the intermediate service variable and from the global upstream components:

$$K_{2,t} \geq \theta_{K_2} \Rightarrow K_{1,t} \geq \theta_{K_1}^{\text{low}} \wedge P_{b,t} \geq \theta_{P_b} \wedge C_{2,t} \geq \theta_{C_2} \\ \wedge D_{2,t} \geq \theta_{D_2} \wedge Q_t \geq \theta_Q \wedge L_t \geq \theta_L \wedge M_t \leq \theta_M.$$

The threshold values used for these conditions are reported in [Table 9](#).

Table 9: Threshold values for the medium-complexity dependency graph.

Threshold	Value	Threshold	Value
θ_{P_m}	0.55	θ_{P_b}	0.50
θ_Y	0.50	θ_{C_1}	0.60
θ_{C_2}	0.60	θ_{S_1}	0.55
θ_{S_2}	0.50	θ_{D_1}	0.55
θ_{D_2}	0.60	θ_Q	0.60
θ_L	0.50	θ_M	0.40
θ_{K_1}	0.70	θ_{K_2}	0.70
$\theta_{K_1}^{\text{low}}$	0.55		

B.2 Dependency Constraints

Dependency constraints encode consistency between downstream variables and their upstream parents. In the medium-complexity regime, the main dependency caps are specified by the following inequalities:

$$C_{1,t} \leq 0.15 + 0.45Y_t + 0.25P_{m,t} + 0.15Q_t + \eta, \\ C_{2,t} \leq 0.10 + 0.35Y_t + 0.35P_{b,t} + 0.20Q_t + \eta, \\ D_{1,t} \leq 0.10 + 0.40S_{1,t} + 0.30C_{1,t} + 0.20Y_t + \eta, \\ D_{2,t} \leq 0.05 + 0.25D_{1,t} + 0.30S_{2,t} + 0.25C_{2,t} + 0.20Q_t + \eta, \\ Q_t \leq 0.10 + 0.30Y_t + 0.25S_{1,t} + 0.25S_{2,t} + 0.20D_{1,t} + \eta, \\ K_{1,t} \leq \min(P_{m,t}, C_{1,t}, D_{1,t}, Q_t, L_t, 1 - M_t) + \eta, \\ K_{2,t} \leq \min(K_{1,t}, P_{b,t}, C_{2,t}, D_{2,t}, Q_t, L_t, 1 - M_t) + \eta.$$

The dependency slack is fixed at $\eta = 0.05$. For each cap, the corresponding violation margin is the positive excess of the generated downstream value above the right-hand side. These margins are aggregated over all dependency constraints and time steps to obtain the dependency-family violation score.

B.3 Mutual-Exclusion Constraints

Mutual-exclusion constraints prevent high service outputs from co-occurring with maintenance or safe-mode conditions. In the medium-complexity regime, the service variables are constrained as follows:

$$\begin{aligned} M_t \geq 0.60 &\Rightarrow K_{1,t} \leq 0.30, \\ M_t \geq 0.60 &\Rightarrow K_{2,t} \leq 0.30. \end{aligned}$$

The associated violation score is computed from the positive excess above the allowed service level when maintenance or safe-mode conditions are active.

B.4 Temporal-Ordering Constraints

Temporal-ordering constraints require selected downstream activations to be preceded by stable upstream conditions. The stability window is set to $q = 2$. If either K_1 or K_2 becomes high at time t , the corresponding prerequisites must have remained above their thresholds over the interval $\{t - q, \dots, t\}$.

For K_1 , the prerequisite set is

$$\mathcal{P}_{K_1} = \{P_m, C_1, D_1, Q, L\}.$$

For K_2 , the prerequisite set is

$$\mathcal{P}_{K_2} = \{K_1, P_b, C_2, D_2, Q, L\}.$$

Thus, a temporal-ordering violation is recorded when a high-service activation occurs before the relevant prerequisite variables have remained stable throughout the required window.

A second temporal condition concerns recovery after cyber degradation. If cyber integrity falls below 0.35, then C_1 , C_2 , D_2 , and Q are not allowed to recover above their high thresholds before Y recovers above 0.50. The corresponding violation score aggregates the positive recovery excess observed before cyber recovery has occurred.

B.5 Structural Invariants

Structural invariants enforce basic graph-state admissibility independently of service-specific dependencies. They include bounded node and edge values, consistency with the allowed-edge topology, and continuity restrictions over edge-state evolution. Generated node and allowed-edge values are required to remain in $[0, 1]$. Variables corresponding to forbidden edges are not generated as independent state components, consistently with the graph-state formulation in [Section 2.2](#).

Continuity restrictions limit abrupt edge-state changes. Unless topology perturbation is active, the maximum allowed edge jump between consecutive time steps is

$$\Delta_{\max} = 0.25.$$

Structural-invariant violation scores aggregate out-of-range values, allowed-edge inconsistencies, and excess jumps over the generated trajectory.

C Supplementary Reliability Analysis

This appendix presents supplementary analyses that support the reliability assessment presented in Section 3.

The additional results are not used to establish the main claims of the paper. Instead, they provide complementary diagnostics regarding structural admissibility, sample efficiency, calibration behavior, and robustness across the two benchmark regimes.

Consistent with the main manuscript, the supplementary analyses are organized around the three reliability dimensions investigated throughout the study:

1. structural admissibility;
2. sample efficiency;
3. decision-relevant calibration.

C.1 Full Experiment 1 Strategy Metrics

The compact graph regime is summarized in Section 3.1, while the cross-regime summary is reported in Table 3. Table 10 provides the corresponding full strategy-level metrics for Experiment 1.

Table 10: Full strategy-level metrics for the compact-graph regime.

Strategy	Invalid mass	Trajectory CVR	Retained ratio	ESS ratio	Diversity	Brier	ECE
Unconstrained diffusion	0.002996	0.002996	1.000000	1.000000	1.030889	0.578380	0.741424
Hard filtering	0.000000	0.000000	0.997004	0.997004	1.028912	0.577716	0.740841
Soft weighting	0.002715	0.002996	1.000000	0.999930	1.030889	0.578309	0.741360
Optional projection	0.000000	0.000000	1.000000	1.000000	1.030910	0.578201	0.741282

C.2 Soft-Weighting Reduction Summary

Soft weighting is discussed in the main text as a probability-reweighting mechanism rather than as a guarantee of structural admissibility; see Section 3.2. Table 11 summarizes the observed invalid-mass reduction and the effective sample size ratio across the two regimes.

Table 11: Soft-weighting effect across the two graph regimes.

Experiment	Unconstrained invalid mass	Soft invalid mass	Absolute reduction	ESS ratio
Experiment 1: compact	0.002996	0.002715	0.000282	0.999930
Experiment 2: complex	0.155929	0.148807	0.007122	0.998764

C.3 Calibration and Robustness Diagnostics

Calibration is treated as a reliability dimension distinct from structural admissibility; see Sections 3.4 and 4.2. Figure 4 provides supplementary calibration diagnostics for admissible-outcome probability estimates.

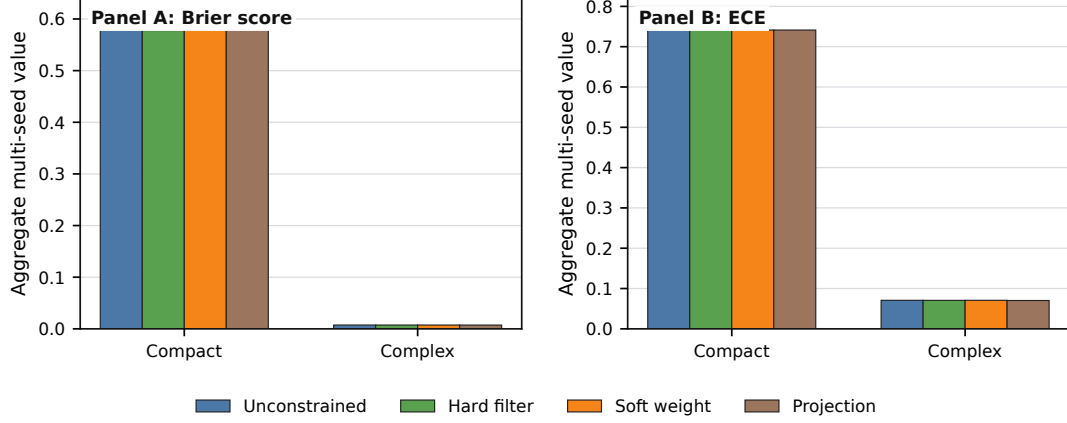


Figure 4: Calibration diagnostics across graph regimes and trajectory-handling strategies. Panel A reports the aggregate multi-seed Brier score, while Panel B reports the aggregate multi-seed expected calibration error (ECE) for admissible-outcome probability estimates.

Robustness diagnostics are discussed in [Section 3.5](#). [Table 12](#) reports the condensed medium-complexity diagnostics under observation-noise and topology-perturbation conditions.

Table 12: Condensed robustness diagnostics in the medium-complexity graph regime.

Condition	Unconstrained invalid mass	Hard retained ratio	Soft invalid mass
Noise base	0.154819	0.845181	0.147811
Noise severe	0.158594	0.841406	0.151528
Topology base	0.156555	0.843445	0.149428
Topology severe	0.169536	0.830464	0.161635

D Reproducibility and Provenance

This appendix documents the provenance of the empirical results and the reproducibility protocol adopted throughout the study.

The full experimental configuration uses seeds 2026, 2027, and 2028 for both benchmark regimes. Because the primary objective of the paper is to evaluate reliability across graph-complexity regimes and constraint-handling strategies, all quantitative claims reported in Section 3 are based exclusively on multi-seed aggregate analyses.

Seed-level outputs are preserved for reproducibility, auditing, traceability, and debugging purposes; however, they are not treated as primary evidence for the scientific claims reported in the manuscript.

Experimental artifacts are organized into compact-regime, medium-complexity-regime, and comparative outputs. The comparative artifacts are derived only after the experiment-level workflows have been completed, preventing cross-regime mixing during training, sampling, metric computation, and figure generation.

The reproducibility package includes configuration files, run manifests, machine-readable aggregate tables, seed-level tables, lambda-sensitivity outputs, near-validity diagnostics defined as fractions of samples below selected violation-score thresholds, figure-data files, hardware and software metadata, and provenance notes. The clean Grace Hopper run manifests and logs constitute the primary computational provenance for the results reported in the manuscript. Historical or exploratory logs may be retained for traceability, but they are not treated as primary evidence for the aggregate results reported in the paper.

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